Jiawei Xu

Ph.D. Student in Information Studies

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EDUCATION

Ph.D. Student in Information Studies Aug. 2023–Present
iSchool (Information), School of Computing, UT Austin
Committee: Prof. Ying Ding (Chair), Prof. Yan Zhang, Prof. Kenneth Fleischmann
Focus: LLM-Based Agents, Agent Workflow Orchestration, Healthcare AI, Scientific Collaboration

M.S. in Information Science Sep. 2021–Jul. 2023
Peking University
Advised by: Dr. Yi Bu and Dr. Win-bin Huang
Award: Excellent Graduate of Beijing

B.M. in Information Management & Systems Sep. 2017–Jul. 2021
Peking University

PUBLICATIONS

Conference Papers

1. **[COLM 2026] Xu, J.***, Zhai, Z.*, Fang, J., Xu, R., Lu, M., Zhang, Y., Wang, G.†, Chen, T.†, & Ding, Y.†. (2026). SEER: Long-Context Reasoning via Selective Visual-Text Compression. *Conference on Language Modeling (COLM 2026).* [Paper] [Code] *(Acceptance rate: 29%)*
2. **[EMNLP 2025]** Pandit, S., **Xu, J.**, Hong, J., Wang, Z., Chen, T., Xu, K., & Ding, Y*. (2025). MedHallu: A Comprehensive Benchmark for Detecting Medical Hallucinations in Large Language Models. *EMNLP 2025 Main Conference.* [Link] *(Acceptance rate: 22%)*
3. **[AAAI 2025]** Cox, K., **Xu, J.**, Han, Y., Xu, R., Chen, T., Gerych, W., & Ding, Y*. (2025). Mapping from Meaning: Addressing the Miscalibration of Prompt-Sensitive Language Models. In *Proceedings of AAAI 2025.* [Code] [Link] *(Acceptance rate: 23%)*
4. **[ASIS&T 2025] Xu, J.**, Chen, J., Ye, Y., Sembay, Z., Thaker, S., Payne-Foster, P., Chen, J., & Ding, Y*. (2025). Interactive Graph Visualization and Teaming Recommendation in an Interdisciplinary Project's Talent Knowledge Graph. *ASIS&T 2025.* [Demo] *(Acceptance rate: 28%)*
5. **[JCDL 2024] Xu, J.**, Xie, Q., Liu, M., Sembay, Z., Thaker, S., Payne-Foster, P., Chen, J. Y., & Ding, Y*. (2024). Decoding Patterns of Data Generation Teams for Clinical and Scientific Success: Insights from the Bridge2AI Talent Knowledge Graph. In *Proceedings of JCDL 2024. (Acceptance rate: 30%)*
6. **[WWW 2026]** Lu, M., Wu, M., **Xu, J.**, Li, W., Liu, F., Ding, Y., Sun, Y., Lu, J., & Zhang, Y. (2026). From Newborn to Impact: Bias-Aware Citation Prediction. *The Web Conference 2026.* [arXiv]
7. **[ICHI 2025]** Cox, K.*, Qu, G.*, Hsu, C.-Y., **Xu, J.**, Zhou, Y., Tan, Z., Hu, M., Chen, T., Hu, Z., Zhao, Z.†, & Ding, Y.† (2025). Thought Graph: Balancing specificity and uncertainty in LLM-based gene set annotation. *IEEE ICHI 2025.*
8. **[ML4H 2024]** Naeem, A., Li, T., Liao, H.-R., **Xu, J.**, Mathew, A. M., Zhu, Z., Tan, Z., Jaiswal, A. K., Salibian, R. A., Hu, Z., Chen, T., & Ding, Y*. (2024). Path-RAG: Knowledge-Guided Key Region Retrieval for Open-ended Pathology Visual Question Answering. *ML4H 2024.* [Code] [Link]
9. **[WWW 2024]** Hsu, C.-Y., Cox, K., **Xu, J.**, Tan, Z., Zhai, T., Hu, M., Pratt, D., Chen, T., Hu, Z., & Ding, Y*. (2024). Thought Graph: Generating Thought Process for Biological Reasoning. In *Companion Proceedings of the ACM Web Conference 2024 (WWW'24 Companion).* [Link]

Journal Articles

10. [**JASIST**] **Xu, J.**, Zheng, Z., Min, C., Huang, W., & Bu, Y*. (2025). Knowledge Integration and Diffusion Structures of Interdisciplinary Research: A Large-Scale Analysis Based on Propensity Score Matching. *Journal of the Association for Information Science and Technology*. [Code] [PDF]

11. [**ACM HEALTH**] Xu, H., Yi, S., Lim, T., **Xu, J.**, Well, A., Mery, C., Zhang, A., Zhang, Y., Ji, H., Pingali, K., Leng, Y., & Ding, Y*. (2026). TAMA: A Human-AI Collaborative Thematic Analysis Framework Using Multi-Agent LLMs for Clinical Interviews. *ACM Transactions on Computing for Healthcare*. [DOI]

12. [**Scientific Data**] Xu, J., Yu, C., **Xu, J.**, Torvik, V. I., Kang, J., Sung, M., Song, M., Bu, Y., & Ding, Y*. (2025). PubMed Knowledge Graph 2.0: Connecting Papers, Patents, and Clinical Trials in Biomedical Science. *Scientific Data*.

Workshop Proposals

13. [**NeurIPS 2025**] **Xu, J.**, Xiang, T., Rajpurkar, P., Hong, J., Chen, C., Adeli, E., Li, X., Pavlakos, G., Delp, S., Fei-Fei, L., & Ding, Y. (2025). The Second Workshop on GenAI for Health: Potential, Trust, and Policy Compliance. *NeurIPS 2025 Workshop Proposal. (Accepted)*

Working Papers

14. **Xu, J.**, Koesdwiady, A., Bei, S., Han, Y., Huang, B., Wang, D., Chen, Y., Wang, Z., Wang, P., Li, P., & Ding, Y. (2026). Rethinking the Value of Multi-Agent Workflow: A Strong Single Agent Baseline. *arXiv preprint*. [arXiv]

15. **Xu, J.**, Ye, Y., Han, Y., Wang, H., Shomer, H., Li, W., Lu, M., Wu, M., Liu, F., Zhang, Y., Sun, Y., Tang, J., & Ding, Y. (2026). MATRIX: Multi-Agent Teaming Recommendation through EXpertise Gap Identification. ***Under review***

16. **Xu, J.**, Lee, Y., Youssef, A. E., Yun, E., Huang, T., Guo, T., Saber, H., Ying, R., & Ding, Y*. (2025). Beyond Feature Importance: Feature Interactions in Predicting Post-Stroke Rigidity with Graph Explainable AI. *arXiv preprint*. [arXiv]

17. **Xu, J.** (co-first), Raza, Z. (co-first), Lim, T., Boddy, L., Mery, C., Well, A., & Ding, Y*. (2025). LLM-TA: An LLM-Enhanced Thematic Analysis Pipeline for Transcripts from Parents of Children with Congenital Heart Disease. *GenAI4Health Workshop at AAAI*. [Code] [Link]

EMPLOYMENT

Applied Scientist Intern May 2026–Aug. 2026
Amazon, Seattle, WA
Mentors: Dihan Dai and Keyang He; Manager: Jinhe Shi; Technical Advisors: Jiajian Lu and Olcay Boz
Project: Self-improving automated research loop for machine learning tasks
Return intern offer

Applied Scientist Intern May 2025–Nov. 2025
Amazon Search, Palo Alto, CA (SFO24)
Mentors: Arief Koesdwiady, Sisong Bei, Yan Han; Manager: Yutong Chen
Project: Rethinking the Value of Multi-Agent Workflow: A Strong Single Agent Baseline (featured as the cover paper in DAIR.AI Top AI Papers of the Week)
Return intern offer

Graduate Research Assistant Aug. 2023–Aug. 2026
iSchool (Information), School of Computing, UT Austin, Austin, TX

- **Patient Simulator**: Developed a voice-based AI patient simulator to help medical students practice clinical conversations and patient communication. [Demo] Related exploration: SEER, long-context reasoning through selective visual-text compression.
- **Cell Maps for AI (CM4AI)**: Developed interactive knowledge graph visualization and teaming recommendation tools to help researchers explore expertise and identify potential collaborators. [Demo]
- **LLM uncertainty quantification and hallucination detection for health**: Contributed to prompt sensitivity and uncertainty calibration in Mapping from Meaning, and medical hallucination detection benchmarking in MedHallu.
- **LLM-assisted thematic analysis**: Developed methods for identifying themes in interviews with parents of children with congenital heart disease, with evaluation against human analysis and expert review. [LLM-TA]

SERVICE

Workshop Organizer (Student Lead) 2026
GenAI4Health Workshop at NeurIPS 2026, Sydney, Australia

Workshop Organizer (Lead) 2025
GenAI4Health Workshop at NeurIPS 2025, San Diego, CA, USA
Featured speakers: Suchi Saria (JHU), Eric Topol (Scripps), James Zou (Stanford), Jimeng Sun (UIUC), Serena Yeung (Stanford), Diyi Yang (Stanford), Fei Wang (Cornell), Sharon Li (UW-Madison), Tianlong Chen (UNC), et al.

Onsite Student Organizer 2024
NeurIPS 2024 GenAI4Health Workshop, Vancouver, Canada
Featured speakers: James Zou (Stanford), David Sontag (MIT), Zachary Lipton (CMU), Bo Li (UChicago), Su-In Lee (UW), Sanmi Koyejo (Stanford), Tanveer Syeda-Mahmood (IBM), Daguang Xu (NVIDIA), et al.

Academic Service

Journal Reviewer

- ACM Computing Surveys
- Data and Information Management
- Journal of the Association for Information Science and Technology
- Nature Humanities & Social Sciences Communications
- Online Information Review
- Data Intelligence
- Information & Culture
- Journal of Clinical and Translational Science
- Scientometrics

Conference PC Member & Reviewer

- ICML
- NeurIPS
- ACL Rolling Review
- ML4H
- AAAI

INVITED TALKS AND TEACHING EXPERIENCE

AI Agents over Biomedical Knowledge Graphs: Finding Collaborators and Querying Data from Research Papers
NSF Convergence Accelerator meeting Oct. 2026
[Project demo]

CM4AI Talent Knowledge Graph
Virtual Talk Oct. 2024; Apr. 2025

Beyond Correlation: What Factors Influence Scientific Performance
Virtual Talk Aug. 2023

Guest Lecture: LLM Prompt Engineering and Multi-Agent Systems
The University of Texas at Austin 2024–2025

Teaching Assistant: Data Visualization
Peking University 2021–2023

Teaching Assistant: Complex Networks
Peking University 2022

PROFESSIONAL SKILLS

LLM	LLM Agents, Post-training (SFT, RLHF, GRPO), veRL, Ray, Slurm, vLLM
LLM Frameworks	LangChain, Vector Databases (Pinecone, FAISS), RAG
ML & Data	Python, PyTorch, Hugging Face (Transformers, Datasets, PEFT), NumPy, Pandas
Web Dev	JavaScript, Node.js, npm, Svelte, Streamlit
Cloud	AWS, Microsoft Azure
Languages	English (Professional), Mandarin (Native)